

Skills Summary

Solving Linear Systems by Substitution

Use this reference while completing your worksheet or when studying.

1. One variable is already isolated

The method: substitution replaces one variable by an *equal expression*, turning two equations in two unknowns into **one** equation in one unknown.

(1) Find the equation solved for a variable. (2) Put that expression — in parentheses — into the *other* equation.

(3) Solve for the one remaining variable. (4) Back-substitute to get the partner. (5) Check in the untouched equation.

Example: $y = x + 2$ and $2x + y = 11$

Step 1. The first equation already says what y equals, so substituting it into the second leaves an equation in x alone — no isolating work needed.

Step 2. $2x + (x + 2) = 11 \Rightarrow 3x + 2 = 11 \Rightarrow x = 3$; then $y = 3 + 2 = 5$. Check: $2(3) + 5 = 11$
✓ Solution: **(3, 5)**

Try it: $y = 2x - 1$ and $3x + y = 19$

check: **(4, 7)**

2. Isolate a variable first (isolate-then-substitute)

Rule: when nothing is isolated, isolate the variable whose coefficient is 1 or -1 — that choice keeps fractions out of the substitution.

From $x + y = 9$: $y = 9 - x$. Substituting a subtraction needs parentheses: $2x - (9 - x) = 2x - 9 + x$.

Example: $x + y = 9$ and $2x - y = 3$

Step 1. Neither equation is solved for a variable, and y in the first has coefficient 1, so isolating there is the fraction-free choice.

Step 2. $y = 9 - x$, so $2x - (9 - x) = 3 \Rightarrow 3x - 9 = 3 \Rightarrow x = 4$; then $y = 9 - 4 = 5$. Check: $2(4) - 5 = 3$ ✓ Solution: **(4, 5)**

Try it: $x + 2y = 11$ and $3x - y = 5$

check: **(3, 4)**

3. When both variables disappear (special-case)

Read what is left behind:

$-6 = 1$ or $12 = 5$ (**false**) $\Rightarrow$ **no solution**; the lines are parallel.

$8 = 8$ or $0 = 0$ (**true**) $\Rightarrow$ **infinitely many solutions**; the two equations describe the same line.

Losing the variables is information, not a mistake — never write “ $x = 0$ ”.

Example: $2x + y = 6$ and $4x + 2y = 5$

Step 1. Nothing is isolated, so isolate $y = 6 - 2x$ and substitute — then read whatever statement survives.

Step 2. $4x + 2(6 - 2x) = 5 \Rightarrow 4x + 12 - 4x = 5 \Rightarrow 12 = 5$, which is false. No pair can work: no solution (parallel lines).

Try it: $y = 3x - 4$ and $6x - 2y = 8$. Substitute, then say what the surviving statement means.

check: infinitely many solutions

4. Building the system from a story (word-model)

Two unknowns need two equations. Name both unknowns with units, then write one equation per *kind* of total in the story:

a **count** equation (how many things) and a **value** equation (money, acid, length). Solve the count equation for a variable — its coefficients are 1 — and substitute into the value equation.

Example: 12 coins, dimes and quarters, worth 255 cents. How many of each?

Step 1. Let x be dimes and y quarters: the count total gives one equation and the money total the other, so substitution starts from the count equation.

Step 2. $x + y = 12 \Rightarrow x = 12 - y$; then $10(12 - y) + 25y = 255 \Rightarrow 120 + 15y = 255 \Rightarrow y = 9$, so $x = 3$. Dimes and quarters: **(3, 9)**

Try it: 15 tickets, adult at 6 dollars and student at 4 dollars, worth 76 dollars. Find (x, y) with x adult tickets.

check: (2, 8)